INTEREST RATE FORWARD AND FUTURES PRICE DIFFERENCES

Learning Outcome

- explain why forward and futures prices differ

The short maturity of most futures contracts and the ability of most market participants to borrow near risk-free rates for these maturities typically results in little to no distinction between futures and forward prices. An exception to this is the so-called **convexity bias**, which arises given the difference in price changes for interest rate futures versus forward contracts, as illustrated in the following example.

EXAMPLE 5

Interest Rate Forwards versus Futures

Let us return to an example from the prior lesson with an interest rate futures contract of $1,000,000 notional for three-month MRR of 2.21% for one quarter (or 90/360 days). Recall that the underlying deposit contract value was:

$1,005,525 = $1,000,000 × [1 + (2.21% / 4)].

The contract BPV was shown to be $25 (= $1,000,000 × 0.01% × [1/4]).

Consider in contrast a $1,000,000 notional FRA on three-month MRR in three months' time with an identical 2.21% rate. The net payment on the FRA is based upon the difference between MRR and the implied forward rate (IFR):

Net Payment = $(MRR_{B-A} - IFR_{A,B-A})$ × Notional Principal × Period.

For example, if the observed MRR in three months is 2.22% (+0.01%), the net payment *at maturity* would be $25 (= $1,000,000 × 0.01% × [1/4]). However, the settlement of an FRA is based upon the present value of the final cash flow discounted at MRR, so:

Cash Settlement (PV): $24.86 = $25 /(1 + 0.0222 / 4).

If we increase the magnitude of the MRR change at settlement and compare these changes between a long interest rate futures position and a short receive-fixed (pay floating) FRA contract, we arrive at the following result:

$MRR_{3m,3m}$	Short FRA Cash Settlement (PV)	Long Futures Settlement
2.01%	$497.50	$500
2.11%	$248.69	$250
2.21%	$0	$0
2.31%	($248.56)	($250)
2.41%	($497.01)	($500)

Although the settlement values differ due to different conventions across these instruments, note that while the futures contract has a fixed linear payoff profile for a given basis point change, the FRA settlement does not.

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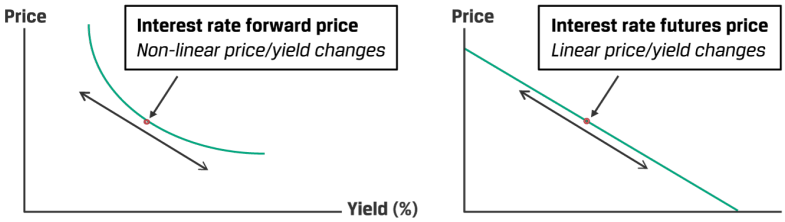
small due to the short forward period, note that this non-linear relationship is the convexity property, which characterizes fixed-income instruments from earlier lessons, as shown in Exhibit 5.

Exhibit 5:

Convexity Bias



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The discounting feature of the FRA, which is not present in the futures contract, leads to a convexity bias that is greater for longer discounting periods. You will recall from an earlier lesson that a discount factor is the *price* equivalent of a zero rate and is the present value of a currency unit on a future date, which may also be interpreted as the price of a zero-coupon cash flow or bond. We will show later how this discount factor is used to price interest rate swaps and other derivatives.

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